1. Who first observed and coined the term 'cell' in 1665?
- **(A)** Robert Hooke
- **(B)** Antonie van Leeuwenhoek
- **(C)** Robert Brown
- **(D)** Theodor Schwann

2. Which model proposed by Singer and Nicolson describes membrane structure?
- **(A)** Unit Membrane Model
- **(B)** Fluid Mosaic Model
- **(C)** Sandwich Model
- **(D)** Lattice Model

3. What is the primary lipid component forming the bilayer of biological membranes?
- **(A)** Triacylglycerols
- **(B)** Phospholipids
- **(C)** Cholesterol
- **(D)** Glycolipids

4. Which organelle contains hydrolytic enzymes active at acidic pH?
- **(A)** Peroxisome
- **(B)** Lysosome
- **(C)** Glyoxysome
- **(D)** Endosome

5. Which molecule serves as a membrane fluidity buffer in animal cell membranes?
- **(A)** Ergosterol
- **(B)** Stigmasterol
- **(C)** Cholesterol
- **(D)** Hopanoid

6. What is the primary site of ribosomal RNA (rRNA) transcription and ribosome assembly?
- **(A)** Nucleolus
- **(B)** Centrosome
- **(C)** Kinetochore
- **(D)** Nuclear lamina

7. Which cellular structure maintains nuclear shape and anchors chromatin?
- **(A)** Nuclear envelope
- **(B)** Nuclear lamina
- **(C)** Pore complex
- **(D)** Nucleoplasm

8. Which transport mechanism moves water across a semipermeable membrane down its potential?
- **(A)** Active transport
- **(B)** Osmosis
- **(C)** Group translocation
- **(D)** Pinocytosis

9. In the Na+/K+ ATPase pump, how many Na+ and K+ ions are transported per ATP hydrolyzed?
- **(A)** 3 Na+ out, 2 K+ in
- **(B)** 2 Na+ out, 3 K+ in
- **(C)** 3 Na+ in, 2 K+ out
- **(D)** 1 Na+ out, 1 K+ in

10. Aquaporins are specialized channel proteins that facilitate the rapid transport of:
- **(A)** Sodium ions
- **(B)** Glucose
- **(C)** Water molecules
- **(D)** Amino acids

11. Which organelle is primarily responsible for the synthesis of cellular lipids and detoxification?
- **(A)** Rough ER
- **(B)** Smooth ER
- **(C)** Golgi Apparatus
- **(D)** Peroxisome

12. Proteins destined for secretion or lysosomal targeting are synthesized on:
- **(A)** Free ribosomes
- **(B)** Rough ER ribosomes
- **(C)** Mitochondrial ribosomes
- **(D)** Peroxisomes

13. What is the major carbohydrate polymer present in bacterial cell walls?
- **(A)** Cellulose
- **(B)** Chitin
- **(C)** Peptidoglycan
- **(D)** Glycogen

14. Which membrane-bound organelle contains catalase to break down hydrogen peroxide?
- **(A)** Lysosome
- **(B)** Peroxisome
- **(C)** Ribosome
- **(D)** Vacuole

15. The resting membrane potential of typical animal cells is predominantly determined by:

(A) Na+ influx
(B) K+ leak channels
(C) Ca2+ ATPase
(D) Cl- active transport

16. Which transport process does NOT require metabolic energy (ATP)?

(A) Primary active transport
(B) Facilitated diffusion
(C) ABC transporter
(D) Receptor endocytosis

17. Movement of a solute against its electrochemical gradient powered by an ion gradient is:

(A) Simple diffusion
(B) Primary active transport
(C) Secondary active transport
(D) Passive filtration

18. The SGLT-1 transporter in intestinal epithelial cells couples the import of glucose to:

(A) Inward Na+ gradient
(B) Outward K+ gradient
(C) Proton gradient
(D) ATP hydrolysis

19. Which junctional complex provides tight intercellular barriers preventing paracellular leakage?

(A) Desmosomes
(B) Gap junctions
(C) Tight junctions (Zonula occludens)
(D) Adherens junctions

20. Connexin proteins assemble into hexameric hemichannels called connexons in:

(A) Tight junctions
(B) Gap junctions
(C) Hemidesmosomes
(D) Focal adhesions

21. The phospholipid bilayer is impermeable to which of the following substances without transporters?

(A) O2 gas
(B) CO2 gas
(C) Steroid hormones
(D) Hydrated Na+ ions

22. Which of the following describes a transmembrane protein with seven alpha-helical domains?

(A) Integrin
(B) GPCR
(C) Cadherin
(D) Selectin

23. Gram-negative bacteria possess which unique outer membrane component?

(A) Teichoic acid
(B) Lipopolysaccharide (LPS)
(C) Mycolic acid
(D) Pseudomurein

24. The nucleosome core particle consists of an octamer containing two copies each of:

(A) H1, H2A, H2B, H3
(B) H2A, H2B, H3, H4
(C) H1, H3, H4, H5
(D) H2A, H2B, H1, H4

25. Which enzyme synthesizes ATP driven by a transmembrane proton motive force?

(A) Na+/K+ ATPase
(B) F0F1 ATP synthase
(C) Ca2+ ATPase
(D) Phosphofructokinase

26. Phagocytosis is a specialized form of endocytosis primarily performed by:

(A) Erythrocytes
(B) Macrophage cells
(C) Fibroblasts
(D) Neurons

27. Which coat protein is responsible for vesicle formation from the plasma membrane during endocytosis?

(A) COP I
(B) COP II
(C) Clathrin
(D) Caveolin

28. Vesicle fusion with a target membrane requires specific pairing between:

(A) Cadherins and Catenins
(B) v-SNARE and t-SNARE
(C) Selectins and Integrins
(D) Tubulin and Kinesin

29. Which lipid is predominantly localized to the outer leaflet of the plasma membrane?

(A) Phosphatidylserine
(B) Phosphatidylcholine
(C) Phosphatidylethanolamine
(D) Phosphatidylinositol

30. Exposure of phosphatidylserine on the outer membrane leaflet serves as an 'eat-me' signal for:

(A) Mitosis
(B) Autophagy
(C) Apoptosis
(D) Endocytosis

31. FRAP (Fluorescence Recovery After Photobleaching) is an analytical technique used to measure:

(A) Protein molecular weight
(B) Lateral diffusion rate of lipids/proteins
(C) Membrane potential
(D) Enzyme kinetics

32. The patch-clamp technique developed by Neher and Sakmann measures:

(A) Single ion channel currents
(B) Vesicle diameter
(C) Osmotic pressure
(D) Nuclear transport rate

33. Which organelle maintains the lowest luminal pH inside a eukaryotic cell (~4.5 to 5.0)?

(A) Endoplasmic reticulum
(B) Golgi cisternae
(C) Lysosome
(D) Mitochondrial matrix

34. ABC transporters utilize energy directly from:

(A) GTP hydrolysis
(B) ATP binding and hydrolysis
(C) Electrochemical H+ gradient
(D) Light absorption

35. The GLUT1 transporter facilitates glucose uptake across human erythrocyte membranes via:

(A) Uniport mechanism
(B) Symport mechanism
(C) Antiport mechanism
(D) Group translocation

36. Which small GTPase regulates nucleocytoplasmic transport through the nuclear pore complex?

(A) Ras
(B) Rho
(C) Ran
(D) Rab

37. Nuclear localization signals (NLS) on cargo proteins are recognized by soluble receptors called:

(A) Importins
(B) Exportins
(C) Karyopherins
(D) Invertases

38. Which organelle contains cristae that dramatically increase its inner membrane surface area?

(A) Chloroplast
(B) Mitochondria
(C) Peroxisome
(D) Lysosome

39. Lipid rafts are specialized microdomains enriched in:

(A) Unsaturated fatty acids
(B) Sphingolipids and cholesterol
(C) Cardiolipin
(D) Phosphatidic acid

40. Cardiolipin (diphosphatidylglycerol) is uniquely abundant in which membrane?

(A) Plasma membrane
(B) Mitochondrial inner membrane
(C) Nuclear envelope
(D) Endosome

41. The Nernst equation calculates the equilibrium potential for an ion. For K+ at 37 deg C, it is:

(A) E_K = 61.5 * log10([K+]out / [K+]in)
(B) E_K = -61.5 * log10([K+]in / [K+]out)
(C) E_K = 26.7 * ln([K+]out / [K+]in)
(D) E_K = RT * ln([K+]out * [K+]in)

42. If [K+]in = 140 mM and [K+]out = 4.5 mM, the Nernst equilibrium potential E_K is approximately:

(A) -91.8 mV
(B) -65.0 mV
(C) +61.5 mV
(D) -20.4 mV

43. The Goldman-Hodgkin-Katz (GHK) voltage equation accounts for relative membrane permeabilities of:

(A) Na+, K+, and Cl-
(B) Ca2+, Mg2+, and H+
(C) Glucose and urea
(D) Proteins and peptides

44. Digitalis and ouabain inhibit which critical transport protein, leading to increased cardiac contractility?

(A) Ca2+ release channel
(B) Na+/K+ ATPase
(C) SGLT2
(D) SERCA pump

45. The multidrug resistance protein 1 (MDR1 / P-glycoprotein) belongs to which transporter superfamily?

(A) Major Facilitator Superfamily (MFS)
(B) ATP-Binding Cassette (ABC)
(C) P-type ATPase
(D) Voltage-gated channel

46. In cystic fibrosis, mutations in CFTR disrupt the ATP-gated transport of which ion?

(A) Na+
(B) K+
(C) Cl-
(D) Ca2+

47. Which class of transport ATPases undergoes a covalent phosphorylated aspartyl intermediate (E1-P/E2-P)?

(A) P-type ATPases
(B) V-type ATPases
(C) F-type ATPases
(D) ABC transporters

48. V-type H+ ATPases are responsible for pumping protons into the lumen of:

(A) Mitochondria
(B) Lysosomes and plant vacuoles
(C) Endoplasmic reticulum
(D) Chloroplast stroma

49. Bafilomycin A1 is a potent, selective pharmacological inhibitor of:

(A) P-type Na+/K+ pump
(B) V-type H+ ATPase
(C) F-type ATP synthase
(D) GLUT4 transporter

50. The rate of simple diffusion of a nonpolar solute across a membrane is governed by:

(A) Michaelis-Menten kinetics
(B) Fick's First Law
(C) Henderson-Hasselbalch equation
(D) Beer-Lambert Law

51. Unlike simple diffusion, facilitated diffusion exhibits:

(A) Linear concentration dependence
(B) Saturation kinetics (Vmax and Km)
(C) Requirement for ATP
(D) Unidirectional transport only

52. The transition temperature (Tm) of a lipid bilayer increases when:

(A) Fatty acid chain length increases
(B) Double bonds (cis-unsaturation) increase
(C) Cholesterol disrupts acyl packing
(D) Phospholipid head groups are removed

53. Which enzyme catalyzes the ATP-dependent translocation of phospholipids from outer to inner leaflet?

(A) Flippase (P4-ATPase)
(B) Floppase (ABC transporter)
(C) Scramblase
(D) Phospholipase C

54. Scramblases facilitate bidirectional lipid movement across the membrane in response to elevated:

(A) Intracellular ATP
(B) Intracellular Ca2+
(C) Extracellular Na+
(D) Luminal H+

55. In the voltage-gated K+ channel (Kv), the voltage-sensing S4 transmembrane segment contains repeats of:

(A) Positively charged Arginine/Lysine
(B) Negatively charged Aspartate
(C) Hydrophobic Leucine residues
(D) Proline kinks

56. The selectivity filter of the KcsA potassium channel coordinates dehydrated K+ ions using:

(A) Backbone carbonyl oxygens
(B) Carboxylate side chains
(C) Amide nitrogens
(D) Water hydration shell

57. Why does the KcsA channel exclude smaller Na+ ions (radius 0.95 A) in favor of K+ (radius 1.33 A)?

(A) Na+ cannot fit into the pore
(B) Desolvation energy of Na+ is not compensated
(C) Na+ binds irreversibly
(D) Na+ carries double charge

58. The ball-and-chain inactivation mechanism of voltage-gated ion channels is also termed:

(A) N-type inactivation
(B) C-type inactivation
(C) Voltage-sensor collapse
(D) Dephosphorylation block

59. Which toxin specifically blocks voltage-gated Na+ channels by occluding the outer pore vestibule?

(A) Tetrodotoxin (TTX)
(B) Tetraethylammonium (TEA)
(C) Curare
(D) Botulinum toxin

60. Tetraethylammonium (TEA) is a classic pharmacological blocker of:

(A) Voltage-gated Na+ channels
(B) Voltage-gated K+ channels
(C) Ca2+ release channels
(D) Chloride channels

61. In epithelial cell transport, paracellular permeability is regulated by claudins located within:

(A) Desmosomes
(B) Tight junctions
(C) Gap junctions
(D) Hemidesmosomes

62. The free energy change Delta G for uncharged solute transport from out to in is given by:

(A) Delta G = RT * ln([C]in / [C]out)
(B) Delta G = RT * ln([C]out / [C]in)
(C) Delta G = z * F * V_m
(D) Delta G = -RT * [C]in

63. For an ion with valence z, the electrochemical free energy change of inward transport is:

(A) Delta G = RT*ln([C]in/[C]out) + z*F*V_m
(B) Delta G = RT*ln([C]out/[C]in) - z*F*V_m
(C) Delta G = z*F*V_m - RT
(D) Delta G = RT / (z*F*V_m)

64. If V_m = -70 mV and z = +1, the electrical contribution (z*F*V_m) for 1 mole of cation entering the cell is:

(A) -6.75 kJ/mol
(B) +6.75 kJ/mol
(C) -14.2 kJ/mol
(D) 0 kJ/mol

65. Which thermodynamic parameter becomes zero when an ion reaches electrochemical equilibrium?

(A) Chemical gradient
(B) Membrane potential
(C) Net electrochemical driving force (Delta G)
(D) Permeability coefficient

66. The sarcoplasmic reticulum Ca2+ ATPase (SERCA) pumps Ca2+ ions from the cytoplasm into the SR lumen at a stoichiometry of:

(A) 1 Ca2+ per ATP
(B) 2 Ca2+ per ATP
(C) 3 Ca2+ per ATP
(D) 4 Ca2+ per ATP

67. Thapsigargin is a specific non-competitive inhibitor of which calcium transporter?

(A) SERCA pump
(B) PMCA pump
(C) IP3 receptor
(D) Ryanodine receptor

68. The gastric proton pump (H+/K+ ATPase) in stomach parietal cells is inhibited by:

(A) Omeprazole
(B) Ouabain
(C) Bafilomycin
(D) Valinomycin

69. Valinomycin is an antibiotic that acts as a mobile carrier ionophore highly selective for:

(A) Na+
(B) K+
(C) Ca2+
(D) Cl-

70. Gramicidin A forms an ion-conducting transmembrane channel that allows passive flow of:

(A) Monovalent cations (H+, Na+, K+)
(B) Divalent cations (Ca2+, Mg2+)
(C) Anions (Cl-, HCO3-)
(D) Neutral sugars

71. The nuclear pore complex (NPC) exhibits an estimated molecular mass of approximately:

(A) 1.2 MDa
(B) 12 MDa
(C) 125 MDa in vertebrates
(D) 500 kDa

72. Transport of molecules larger than ~40 kDa through the nuclear pore complex requires:

(A) Passive diffusion
(B) Ran-GTP dependent active transport
(C) Vesicular budding
(D) Membrane fusion

73. RanGEF (RCC1) converts Ran-GDP to Ran-GTP exclusively inside the:

(A) Cytoplasm
(B) Nucleus
(C) Endoplasmic reticulum
(D) Golgi lumen

74. RanGAP promotes Ran-GTP hydrolysis to Ran-GDP exclusively in the:

(A) Cytoplasm
(B) Nucleus
(C) Nucleolus
(D) Mitochondria

75. The classical nuclear export signal (NES) is characterized by a conserved spacing of:

(A) Basic Arginine residues
(B) Hydrophobic Leucine-rich motifs
(C) Acidic Glutamate residues
(D) Aromatic Tyrosine residues

76. Given [Na+]out = 145 mM and [Na+]in = 12 mM at 37 deg C, the Nernst equilibrium potential for Na+ (E_Na) is:

(A) +66.5 mV
(B) -66.5 mV
(C) +30.2 mV
(D) -90.1 mV

77. In a cell where V_m = -70 mV and E_Na = +66 mV, the electrochemical driving force acting on Na+ ions is:

(A) -136 mV (strong inward pull)
(B) +136 mV (strong outward push)
(C) -4 mV
(D) +4 mV

78. If membrane permeability ratio P_K : P_Na : P_Cl is 1.0 : 0.04 : 0.45, the GHK equation yields a resting V_m closest to:

(A) +55 mV
(B) -70 mV
(C) -10 mV
(D) 0 mV

79. The partition coefficient (K) in membrane permeability relates to the ratio of solute solubility in:

(A) Water to lipid
(B) Olive oil (organic) to water
(C) Blood to cytoplasm
(D) Membrane to protein

80. Permeability coefficient P is defined as (K * D) / Delta_x. What does Delta_x represent?

(A) Solute molecular radius
(B) Membrane hydrocarbon thickness (~4 nm)
(C) Diffusion constant in water
(D) Electric dipole moment

81. The binding of ouabain to the Na+/K+ ATPase stabilizes the enzyme in which conformation?

(A) E1 state with open cytoplasmic gate
(B) Phosphorylated E2-P state facing extracellular
(C) Unphosphorylated E2 state
(D) Occluded E1 state

82. In P-type ATPases, the conserved phosphorylation site containing the aspartate residue has the consensus sequence:

(A) DKTGT
(B) GXXXXGKT
(C) DEAD-box
(D) RGD motif

83. What is the primary thermodynamic driving force enabling the 2Na+/1-glucose symporter (SGLT1) to accumulate intracellular glucose against a 30,000-fold concentration gradient?

(A) ATP hydrolysis at the substrate site
(B) Twofold Na+ electrochemical potential gradient
(C) Membrane hyperpolarization by K+ pump
(D) Phosphorylation of glucose to G6P

84. Calculate the maximum theoretical glucose accumulation ratio [Glucose]in/[Glucose]out for SGLT1 with stoichiometry n=2, assuming Delta mu_Na = -14 kJ/mol at 310 K:

(A) approx 100
(B) approx 1,000
(C) approx 53,000
(D) approx 2

85. In the structural cycle of ABC transporters, ATP binding to the nucleotide-binding domains (NBDs) induces:

(A) NBD dimerization and outward-facing conformation
(B) NBD dissociation and inward-facing conformation
(C) Permanent channel opening
(D) Phosphorylation of the transmembrane domain

86. During vesicular transport, Rab GTPases recruit specific tethering effectors only when bound to:

(A) GDP in cytoplasm
(B) GTP on the vesicle membrane
(C) Inorganic phosphate
(D) SNARE motif

87. The four-helix bundle formed during trans-SNARE pairing consists of which stoichiometric helix contribution?

(A) 1 from v-SNARE, 3 from t-SNARE
(B) 2 from v-SNARE, 2 from t-SNARE
(C) 3 from v-SNARE, 1 from t-SNARE
(D) 4 from v-SNARE alone

88. Disassembly of the post-fusion cis-SNARE complex requires ATP hydrolysis catalyzed by:

(A) Dynamin
(B) NSF (N-ethylmaleimide sensitive factor) and alpha-SNAP
(C) Hsp70 chaperone
(D) Clathrin light chain

89. Dynamin is a large GTPase required during clathrin-mediated endocytosis to:

(A) Form the polyhedral cage
(B) Mechanically pinch off the constricted vesicle neck
(C) Phosphorylate phosphoinositides
(D) Recruit adaptins (AP-2)

90. In the nuclear pore complex, the permeability barrier is constituted by disordered repeat domains rich in:

(A) Lysine and Arginine
(B) Phenylalanine and Glycine (FG repeats)
(C) Proline and Hydroxyproline
(D) Cysteine disulfides

91. Exportin-1 (CRM1) binds its leucine-rich NES cargo cooperatively with which cofactor?

(A) Ran-GDP
(B) Ran-GTP
(C) Importin-beta
(D) ATP-Mg2+

92. The patch-clamp configuration where a tiny patch of membrane is excised with its cytoplasmic face exposed to the bath solution is:

(A) Whole-cell mode
(B) Inside-out patch
(C) Outside-out patch
(D) Cell-attached mode

93. In an outside-out patch clamp recording, the investigator can directly test the effect of:

(A) Extracellularly applied receptor agonists
(B) Intracellular second messengers
(C) Nuclear transcription factors
(D) Cytoplasmic kinases

94. According to the Grahame equation in membrane biophysics, surface charge density creates a Gouy-Chapman:

(A) Linear potential decline
(B) Exponentially decaying electric double layer
(C) Permanent dipole field
(D) Zero field gradient

95. The thermodynamic melting temperature (Tm) of dipalmitoylphosphatidylcholine (DPPC, 16:0) bilayer is approximately:

(A) 0 deg C
(B) +41 deg C
(C) -18 deg C
(D) +85 deg C

96. The critical micelle concentration (CMC) of a surfactant or detergent is the concentration at which:

(A) Monomer solubility drops to zero
(B) Thermodynamically stable polymolecular micelles spontaneously form
(C) Membranes undergo complete lysis
(D) Proteins precipitate irreversibly

97. Which non-ionic detergent has a low CMC (approx 0.2 mM) and is widely used to solubilize integral membrane proteins in native states?

(A) SDS (Sodium Dodecyl Sulfate)
(B) Triton X-100
(C) Urea
(D) Guanidinium chloride

98. Bacteriorhodopsin from Halobacterium salinarum functions as a light-driven:

(A) Chloride pump
(B) Outward proton (H+) pump
(C) Sodium-potassium exchanger
(D) ATP synthase

99. The chromophore retinal in bacteriorhodopsin is covalently linked to Lys-216 via a:

(A) Thioester bond
(B) Protonated Schiff base
(C) Phosphodiester linkage
(D) Disulfide bond

100. During the bacteriorhodopsin photocycle, light absorption triggers photoisomerization of retinal from:

(A) 13-cis to all-trans
(B) all-trans to 13-cis
(C) 11-cis to all-trans
(D) all-trans to 9-cis

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Differentiate between prokaryotic and eukaryotic cellular architecture regarding compartmentalization and genetic material organization.

Q2. Describe the Fluid Mosaic Model of Singer and Nicolson (1972) and explain the biochemical factors influencing membrane fluidity.

Q3. Derive the thermodynamic equation for uncharged solute transport across a biological membrane and calculate Delta G for an inward 10-fold concentration gradient.

Q4. Formulate the electrochemical potential equation for ion transport and explain how the Nernst equilibrium potential is derived.

Q5. Explain the enzymatic mechanism and structural cycle of the P-type Na+/K+ ATPase, detailing the E1 and E2 conformational states.

Q6. Discuss the Goldman-Hodgkin-Katz (GHK) voltage equation and how resting membrane potential is generated and maintained.

Q7. Describe the molecular architecture of voltage-gated potassium channels (Kv) and explain their selectivity filter mechanism.

Q8. Explain the structural organization and functional role of lipid rafts in plasma membrane signaling.

Q9. Contrast simple diffusion, facilitated diffusion, primary active transport, and secondary active transport with biological examples.

Q10. Describe the structure of ABC transporters and detail their role in multidrug resistance (MDR) in cancer chemotherapy.

Q11. Detail the molecular mechanism of SGLT-1 and explain how oral rehydration therapy (ORT) leverages this transport pathway.

Q12. Describe the nuclear pore complex (NPC) architecture and the molecular cycle of Ran-regulated nucleocytoplasmic transport.

Q13. Explain the biophysical principles and experimental execution of the Patch-Clamp technique in measuring single ion channel activity.

Q14. Discuss the biochemical mechanisms of vesicular docking and fusion mediated by SNARE proteins and Rab GTPases.

Q15. Explain the structural basis of Clathrin-mediated endocytosis, detailing the roles of Clathrin triskelions, AP-2, and Dynamin.

Q16. Describe the structure and physiological role of Gap Junctions, explaining their gating mechanisms in cardiac synchronization.

Q17. Explain the biochemical mechanisms by which cells maintain asymmetric phospholipid distribution across the plasma membrane.

Q18. Describe the structural organization of Tight Junctions (Zonula Occludens) and explain their barrier and fence functions in epithelial polarity.

Q19. Discuss the CFTR chloride channel, its domain architecture, gating by ATP, and how the Delta-F508 mutation causes Cystic Fibrosis.

Q20. Detail the structure and catalytic cycle of F0F1 ATP Synthase and explain how the proton motive force drives rotary catalysis.

Q21. Explain the mechanism of primary active transport in the Gastric H+/K+ ATPase and how proton pump inhibitors (PPIs) inhibit acid secretion.

Q22. Discuss the structural organization and function of Desmosomes and Hemidesmosomes in maintaining tissue mechanical integrity.

Q23. Explain the biophysical mechanisms underlying Fluorescence Recovery After Photobleaching (FRAP) to quantify membrane diffusion coefficients.

Q24. Describe the biophysical properties of ionophores, contrasting mobile carriers (Valinomycin) with pore-forming channels (Gramicidin A).

Q25. Explain how the Nernst-Planck electrodiffusion equation models ion flux across biological membranes.

Q26. Discuss the biochemical composition, assembly, and function of the Nuclear Lamina, and explain laminopathies caused by LMNA mutations.

Q27. Describe the molecular mechanisms of primary cilia as cellular antennae and the role of intraflagellar transport (IFT).

Q28. Explain the structural mechanics of the bacterial flagellar motor as a rotary nanomachine driven by proton flux.

Q29. Discuss the molecular composition of the Extracellular Matrix (ECM) and explain how Integrins mediate bidirectional inside-out and outside-in signaling.

Q30. Explain the mechanism of Autophagy, detailing the molecular stages of autophagosome nucleation, elongation, and lysosomal clearance.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[D]	**Q41:**	[A]	**Q61:**	[B]	**Q81:**	[B]
Q2:	[B]	**Q22:**	[B]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[B]	**Q23:**	[B]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[B]
Q4:	[B]	**Q24:**	[B]	**Q44:**	[B]	**Q64:**	[A]	**Q84:**	[C]
Q5:	[C]	**Q25:**	[B]	**Q45:**	[B]	**Q65:**	[C]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[B]	**Q46:**	[C]	**Q66:**	[B]	**Q86:**	[B]
Q7:	[B]	**Q27:**	[C]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[B]	**Q28:**	[B]	**Q48:**	[B]	**Q68:**	[A]	**Q88:**	[B]
Q9:	[A]	**Q29:**	[B]	**Q49:**	[B]	**Q69:**	[B]	**Q89:**	[B]
Q10:	[C]	**Q30:**	[C]	**Q50:**	[B]	**Q70:**	[A]	**Q90:**	[B]
Q11:	[B]	**Q31:**	[B]	**Q51:**	[B]	**Q71:**	[C]	**Q91:**	[B]
Q12:	[B]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[B]	**Q92:**	[B]
Q13:	[C]	**Q33:**	[C]	**Q53:**	[A]	**Q73:**	[B]	**Q93:**	[A]
Q14:	[B]	**Q34:**	[B]	**Q54:**	[B]	**Q74:**	[A]	**Q94:**	[B]
Q15:	[B]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[B]	**Q95:**	[B]
Q16:	[B]	**Q36:**	[C]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[B]
Q17:	[C]	**Q37:**	[A]	**Q57:**	[B]	**Q77:**	[A]	**Q97:**	[B]
Q18:	[A]	**Q38:**	[B]	**Q58:**	[A]	**Q78:**	[B]	**Q98:**	[B]
Q19:	[C]	**Q39:**	[B]	**Q59:**	[A]	**Q79:**	[B]	**Q99:**	[B]
Q20:	[B]	**Q40:**	[B]	**Q60:**	[B]	**Q80:**	[B]	**Q100:**	[B]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

3. Directionality Criteria:
- When [C]out > [C]in, [C]in / [C]out < 1, ln([C]in / [C]out) < 0, so Delta G < 0 (Spontaneous passive inward diffusion).
- When [C]in > [C]out, [C]in / [C]out > 1, ln([C]in / [C]out) > 0, so Delta G > 0 (Endergonic; requires coupled energy).
4. Calculation for 10-fold Inward Gradient ([C]out / [C]in = 10, so [C]in / [C]out = 0.10) at 310 K (37 deg C):
Delta G = (8.314 J/mol*K) * (310.15 K) * ln(0.10) = 2,578.6 J/mol * (-2.3026) = -5,937.5 J/mol = -5.94 kJ/mol.
Interpretation: Solute entry releases 5.94 kJ/mol of free energy, driving spontaneous downhill uptake.

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nucleotide site accelerates transition back to the E1 state, releasing 2 K+ into the cytoplasm.

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1. Definition and Composition: Lipid rafts are dynamic nanoscale (10-200 nm), sterol- and sphingolipid-enriched microdomains within the plasma membrane. Sphingolipids have long, saturated acyl chains that pack tightly in a liquid-ordered (Lo) phase, segregated from the surrounding fluid liquid-disordered (Ld) glycerophospholipid matrix.
2. Physicochemical Properties: Rafts are insoluble in cold non-ionic detergents (e.g., 1% Triton X-100 at 4 deg C), earning the operational definition Detergent-Resistant Membranes (DRMs). Cholesterol acts as molecular glue filling voids between bulky sphingolipid head groups.
3. Functional Roles:
- Spatial Organization of Signal Transduction: Rafts selectively recruit GPI-anchored proteins, palmitoylated/myristoylated peripheral proteins (e.g., Src family kinases like Lck and Fyn), while excluding transmembrane phosphatases (e.g., CD45). Receptor clustering inside rafts triggers localized kinase phosphorylation cascades.
- Platform for Pathogen Entry: Viruses (HIV, Influenza) and bacterial toxins (Cholera toxin binding to ganglioside GM1) exploit rafts for receptor docking and cell entry.

>>**Solution to Question 30:** Contrast the diffusion-facilitated diffusion and their active transport and resistance (MDR) transporter with biological examples.

Substrate binds to the high-affinity inward-facing cavity. Two ATP molecules bind at the interface between the two NBDs, inducing NBD dimerization. Dimerization induces outward rigid-body tilting of TMDs, releasing the substrate outside. Subsequent ATP hydrolysis and inorganic phosphate release resets the NBD dimer back to open monomeric state.

3. Multidrug Resistance (MDR1 / P-glycoprotein):

P-gp (ABCB1) is overexpressed on cancer cell membranes following exposure to chemotherapeutics. It recognizes diverse hydrophobic, amphipathic anticancer drugs (doxorubicin, paclitaxel, vincristine) directly from the inner membrane leaflet, pumping them out before reaching nuclear DNA targets, causing broad cross-resistance.

>>Solution Question 12: Describe the molecular mechanism of SNARE-mediated membrane fusion, including the roles of Rab GTPases and tethering factors.

3. Nuclear Import Cycle:
- Cargo containing classical Nuclear Localization Signal (NLS, e.g., PKKKRKV) binds Importin-alpha/beta in cytoplasm.
- The complex navigates through FG-repeats via transient hydrophobic interactions.
- Inside the nucleus, Ran-GTP binds Importin-beta with high affinity, causing allosteric dissociation and cargo release into nucleoplasm.
- Importin-beta-Ran-GTP complex exports back to cytoplasm, where RanGAP stimulates hydrolysis to Ran-GDP, recycling Importin.

>>Solution to Question 14: Discuss the biochemical mechanisms of vesicular docking and fusion, including the role of SNARE proteins and Rab GTPases.

displacing hydrating water. The outer leaflets merge first (hemifusion stalk), followed by inner leaflet coalescence to open an aqueous fusion pore.

4. Complex Disassembly and Recycling:

After fusion, the collapsed cis-SNARE complex is disassembled by the hexameric AAA+ ATPase NSF and its adaptor alpha-SNAP, utilizing ATP hydrolysis to liberate monomers for future rounds.

>>Solution Question 18: Describe the structural basis of Clathrin-mediated Endocytosis, detailing the role of Clathrin triskelions, AP2 adaptor complex.

propagate rapidly via gap junction electrotonic current flow without synaptic delay, enabling the entire ventricular syncytium to contract as a coordinated functional unit.

4. Gating Mechanisms:

Channels close rapidly under pathological conditions:

- High intracellular [Ca2+] (> 1 microM) or severe acidosis (pH < 6.5) triggers connexin closure, electrically isolating ischemic or damaged cells to prevent necrosis from spreading to healthy neighbors.

>>Solution to Question 15: Describe the ischemic mechanisms by which cells maintain ion homeostasis and the plasma membrane pol

non-specific passive diffusion of solutes, pathogens, and water between adjacent cells. In the blood-brain barrier, claudin-5 generates high trans-epithelial electrical resistance (> 1,000 Ohm*cm^2).

- The Fence Function: Prevents lateral diffusion of integral membrane proteins and outer-leaflet lipids between the apical domain (rich in microvilli, SGLT1, hydrolytic enzymes) and basolateral domain (rich in Na+/K+ ATPase, GLUT2). This preserves functional apical-basal epithelial polarity essential for directional vector transport.

>>Solution Store Question Bank Structure and dynamics of F0F1-ATP synthase and ex-vivo ATP powered motor Delta F508 mutation causes Cystic Fibrosis.

3. Rotary Catalysis in F1:
Rotation of the asymmetrical gamma shaft within the stationary (alpha-beta)3 head forces each beta catalytic subunit through three sequential conformations:
- Open (O) Conformation: Very low substrate affinity; newly formed ATP dissociates, leaving the site empty.
- Loose (L) Conformation: Binds ADP and inorganic phosphate (Pi) loosely; substrate cannot escape.
- Tight (T) Conformation: Substrates are compressed together with catalytic residues, spontaneously synthesizing ATP without external energy input.
Proton-driven gamma-shaft rotation drives the cooperative transition: T -> O, L -> T, O -> L, producing 3 ATP molecules per complete 360-degree rotation.

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extracellular basement membrane (lamina lucida/densa).
- Transmembrane Receptors: Integrin alpha-6-beta-4 heterodimer and Collagen XVII (BP180). Integrin alpha-6-beta-4 binds laminin-332 in the basal lamina.
- Cytoplasmic Plaque: Plectin and BP230 bridge the integrin cytoplasmic tail to the cytokeratin network.
3. Pathological Significance:
- Pemphigus Vulgaris: Autoantibodies against Desmoglein-3 cause loss of intercellular desmosomal adhesion (acantholysis), resulting in severe intraepidermal blistering.
- Bullous Pemphigoid: Autoantibodies against hemidesmosomal BP180/BP230 detach the epidermis from the dermis, causing deep, tense subepidermal blisters.

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2. Channel-Forming Ionophores (Example: Gramicidin A):
- Structure: Linear 15-amino-acid peptide with alternating L- and D-amino acids.
- Mechanism: Two gramicidin molecules in opposing leaflets dimerize head-to-head via 6 intermolecular hydrogen bonds, forming a continuous 0.4 nm diameter cylindrical beta-helix across the membrane.
- Translocation: Ions diffuse rapidly through the aqueous pore at rates up to 10^7 ions/second (1,000 times faster than mobile carriers).
- Temperature Sensitivity: Can conduct ions even below Tm as long as pre-formed dimers remain intact.

>>**Solution to Question 38:** Describe the chemical modification of the Nuclear Lamina and explain laminopathies caused by LMNA muta

- Mechanical scaffolding maintaining nuclear shape and resisting shear stress.
- Chromatin tethering: Binds heterochromatin via Lamina-Associated Domains (LADs), silencing transcription.
- Positioning of nuclear pore complexes.

3. Laminopathies (Hutchinson-Gilford Progeria Syndrome - HGPS):
- A de novo point mutation in LMNA (c.1824C>T, p.Gly608Gly) activates a cryptic splice donor site, deleting 50 amino acids near the C-terminus.
- This removes the ZMPSTE24 cleavage site, resulting in a mutant protein called Progerin that remains permanently farnesylated and membrane-stuck.
- Progerin accumulation disrupts lamina integrity, causes nuclear blebbing, loss of heterochromatin, telomere damage, and accelerated organismal aging.

>>**Solution to Question 25:** Describe the structural mechanism of bacterial flagellar motor assembly and the role of intraflagellar transport (IFT).

- MotA and MotB form two proton-conducting channels. As protons flow from the periplasm through MotA/MotB down the electrochemical gradient, protonation/deprotonation of a critical aspartate (Asp-32 of MotB) drives conformational power strokes in the cytoplasmic domain of MotA.
- MotA interacts electrostatically with charged residues on the C-terminal domain of FliG in the C-ring, rotating the rotor at speeds up to 100,000 rpm with near 100% thermodynamic efficiency.

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proteins (p62/SQSTM1). The phagophore seals into a closed double-membrane autophagosome.
4. Lysosomal Fusion and Degradation:
STX17 (v-SNARE) on the outer autophagosomal membrane pairs with SNAP-29 and VAMP8 (t-SNAREs) on lysosomes, assisted by Rab7. Autolysosome formation exposes cargo to acid hydrolases, degrading macromolecules into amino acids and fatty acids recycled for survival.
